

Solution Brief: Secure Migration to AWS Cloud

Executive Summary

Ascella Technologies is proposing that your agency transition from using on-premise servers to using the Amazon Web Services (AWS) Cloud environment. This transition will eliminate server crashes, improve security, and reduce maintenance. Having assisted multiple other government clients in transitioning to cloud-based environments, we expect to create a near seamless transition for your organization.

The Problem

Your organization had four server crashes last quarter: Three of these crashes resulted from large unexpected spikes in traffic, and one resulted from a hacker gaining unauthorized access to your database with admin privileges. In addition to this, your on-premise servers have high maintenance costs.

The Solution

To eliminate server crashes going forward, to improve security, and to reduce maintenance costs, our organization suggests that you transition to a secure AWS Cloud environment.

Here's the basic steps we would suggest:

- First, implement Terraform for Infrastructure as Code to ensure to reduce maintenance costs.
 - Currently, your agency manually configures its on-site servers. This is particularly time consuming and also poses a significant risk, since human error is possible every time your servers need to be updated. To fix this issue, we suggest using configuration files for your cloud environment. Using Terraform, these could be easily duplicated, modified, and version controlled, eliminating the risks associated with manual configuration.
- Second, move the SQL databases to Amazon RDS for auto-scaling.
 - Unexpected spikes in traffic cause your servers to crash because your servers can only handle so much traffic. By moving your SQL databases to the AWS Cloud and then having Amazon RDS manage auto-scaling, you can eliminate these crashes. (This is because auto-scaling makes it so that, whenever traffic to your databases increases, the amount of computation available to you also scales correspondingly. This way, you always have enough computation no matter how much traffic you experience.) Additionally, since Amazon RDS automatically performs database back-ups, database software patching, and operating system updates, this transition will eliminate a lot of the complexity and maintenance costs that come with having on-premise servers.
- Third, improve security by using IAM roles and VPC encryption.
 - Currently, any user on your database has almost full permissions for accessing and modifying data. Last quarter, this led to a server crash because a hacker was able to access a configuration file that should have required higher permissions to access. To eliminate this issue, our organization will implement IAM roles, which make it so that users and software only have permission to modify or access the data that is necessary for their role. These roles also make it so that every user's actions are logged and so that their permissions can be revoked at any time. This will prevent hackers from being able to

create significant damage if they're only able gain access to a single individual's permissions.

- We will also implement VPC encryption, which allows you to manage the traffic on your network and to ensure that it is properly encrypted.

Comparison Between On-Premise Servers and Our AWS Cloud Solution

	On-Premise Servers	Our AWS Cloud Solution
IT Infrastructure	Requires manual set-up and is prone to human error	Uses IaC and Terraform, allowing environments to be easily created, duplicated, and version controlled, with minimal human effort.
Handling Traffic	Requires enough servers to handle peak traffic even if average traffic is much lower than peak traffic.	With Amazon RDS, computation automatically scales to handle any amount of traffic.
Cybersecurity	Current permissions allow individual users to access data they should not have access to.	IAM roles and VPC encryption ensure that users only have access to the data they should be able to and that all data is encrypted, tracked, and easily managed.
Reliability	Crashes can occur if servers receive unexpected levels of traffic.	Crashes are extremely unlikely as computation scales with traffic and data is backed onto multiple servers in different locations.
Cost	Scales with servers, labor, and space.	Scales with traffic.

Addressing Concerns About Security And Complexity of Transition

We have heard some concerns about the security and complexity of this transition.

There are three important factors to note:

1. Ascella Technologies has been assisting in cloud transitions for over twenty years and has successfully transitioned multiple government clients without any security issues.
2. Although transitions can be complex, the majority of complexity in these transitions comes from legacy software that already needs to be re-modernized to function in a modern software environment.
3. The AWS Cloud offers extremely high levels of security. Currently, AWS Cloud has achieved accreditation for all US government data classifications in regards to security and compliance requirements. As a result, over 7,500 government agencies use AWS, including the DOW, the CIA, and the NSA. Given how many agencies have decided to undergo transitions, we are confident that a transition would be a good choice for your agency as well.